21.10.2024

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Versuch 242

Versuchsaufbau:

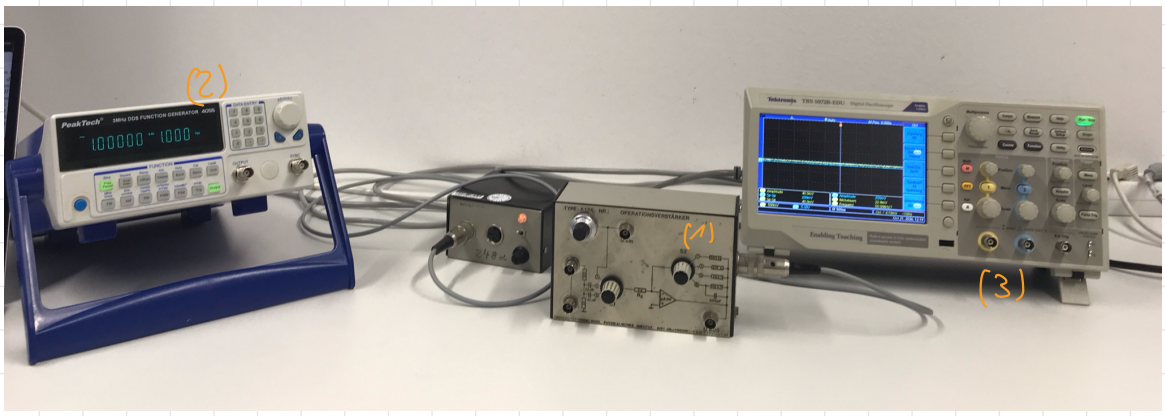

Abb. 3.1 Messaufbau

Geräte:

Operationsverstärker µA 741 (1)

Signalgenerator (2)

Oszilloskop (3)

Aufgabe 1:

Tabelle 3.1: Messung A1

	U_1 [mV]	U_a [V]
1	250,0	-3,89
2	200,0	-3,10
3	150,0	-2,31
4	100,0	-1,40
5	50,0	-0,60
6	0	0,20
7	-50	1,00
8	-100	2,00
9	-150	2,67
10	-200	3,41
11	-250	4,36

$\Delta U_1 = 1,0$ mV

$\Delta U_A = 0,10$ V

48,7 kΩ

Tabelle 3.2

	U_1 [V]	U_a [V]
1	250	-12,80
2	200	-12,80
3	150	-12,70
4	100	-9,20
5	50	-4,74
6	0	-0,12
7	-50	4,42
8	-100	8,81
9	-150	13,80
10	-200	14,60
11	-250	14,70

$\Delta U_1 = 1,0\,mV$

$\Delta U_A = 0,10\,V$

274 kΩ

$f = 1\,kHz$

$(0,05 \leq U_{ss} \leq 1)\,V_{ss}$

274 kΩ

U_G - Effektivwert

$\Delta U_G = 3\,mV$

$\Delta U_a = 10\,mV$

Tabelle 3.3

	U_{ss} [mV_{ss}]	U_G [mV]	U_a [mV]
1	50	14	130
2	200	58	531
3	350	103	921
4	500	146	1320
5	650	190	1710
6	800	235	2110
7	1000	293	2620

Tabelle 3.4:

	U_{ss} [mV_{ss}]	U_G [mV]	U_a [mV]
1	50	14	321
2	200	59	1290
3	350	103	2250
4	500	146	3180
5	650	191	4130
6	800	235	5090
7	1000	293	6370

680 kΩ

$U_e = 10\,U_G$

Aufgabe 2: Phyton

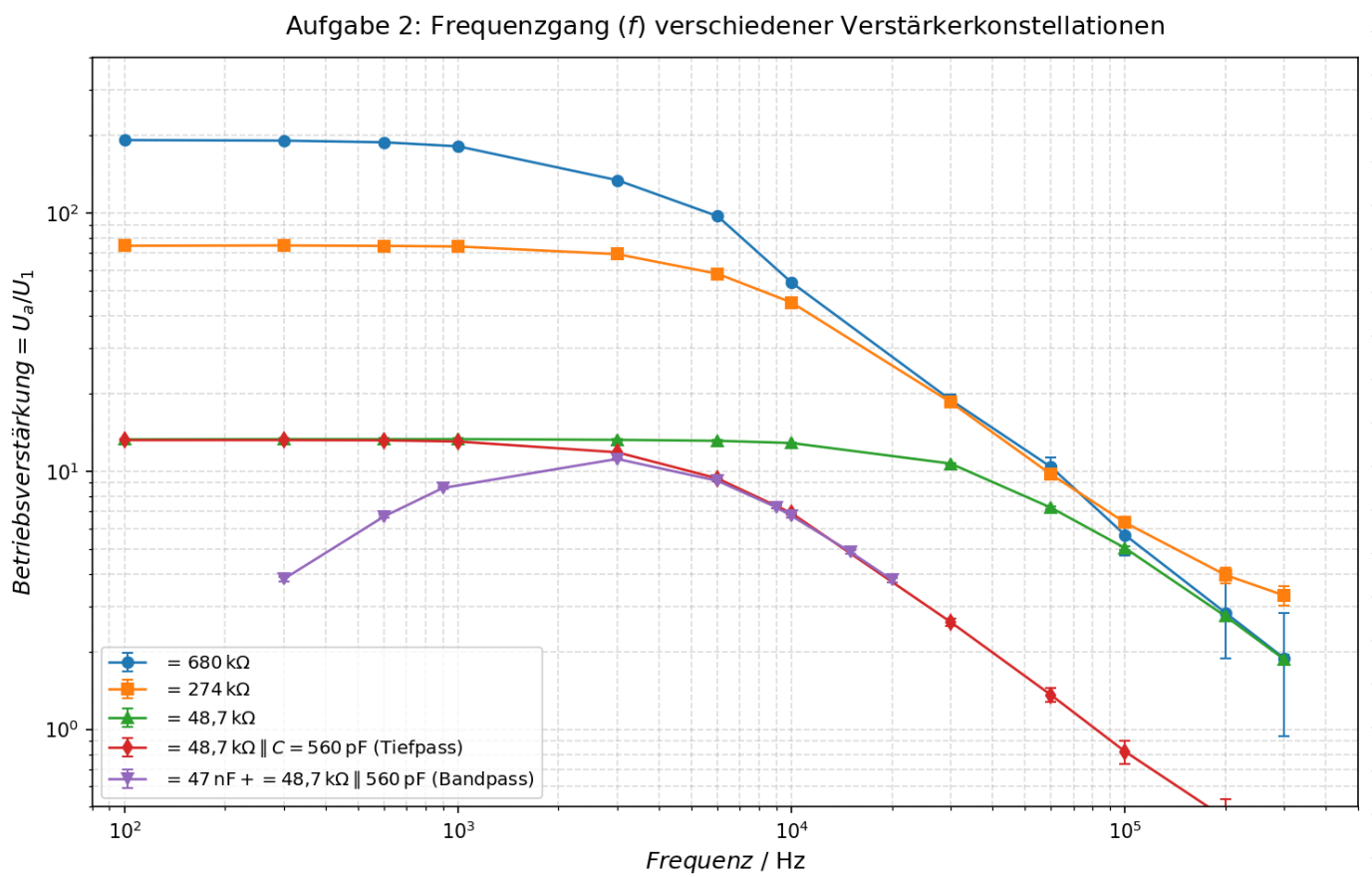

Abb. 3.2: Messwerte

$$\frac{U_{a,eff}}{U_{1,eff}} = \frac{2 \cdot 10\sqrt{2}\, U_{a,eff}}{U_g}$$

```
f1 = np.array([100, 300, 600, 1000, 3000, 6000, 10000, 30000, 60000, 100e3, 200e3, 300e3])
```
✓ 0.0s

```
# 0.3VPP

Ueff_680 = np.array([2.03, 2.02, 1.99, 1.92, 1.42, 1.03, 0.57, 0.20, 0.11, 0.06, 0.03, 0.02])

delta_Ueff_680 = 0.010

Ueff_274 = np.array([791, 793, 790 , 786, 734, 616, 477, 196, 103, 67, 42, 35])*1e-3

delta_Ueff274 = 3e-3


# 1Vpp

Ueff_487 = np.array([469, 470, 470, 470, 467, 463, 454, 378, 255, 178, 97, 66])*1e-3

delta_487 = 3e-3


# 1Vpp, C=560pF

Ueff_487_560 = np.array([466, 466, 465, 460, 418, 331, 243, 92, 48, 29, 16, 11])*1e-3
```
✓ 0.0s

```
f2 = np.array([300, 600, 900, 3000, 6000, 9000, 10000, 15000, 20000])

# Hochpass (C = 47 nF) mit Gegenkopplung Rg = 48.7 kOhm, C = 560 pF (1Vpp)
# Hinweis: Messwerte waren in mV notiert, daher Umrechnung * 1e-3 in V
Ueff_487_560_47 = np.array([135, 236, 304, 394, 324, 256, 237, 172, 134]) * 1e-3
delta_487_560_47 = 3e-3
delta_487_560 = 3e-3

Us = [Ueff_680, Ueff_274, Ueff_487, Ueff_487_560, Ueff_487_560_47]
delta_Us = [delta_Ueff_680, delta_Ueff274, delta_487, delta_487_560, delta_487_560_47]
```
✓ 0.0s

Abb. 3.3 Messwerte

Aufgabe 3

$f = 1\,kHz$

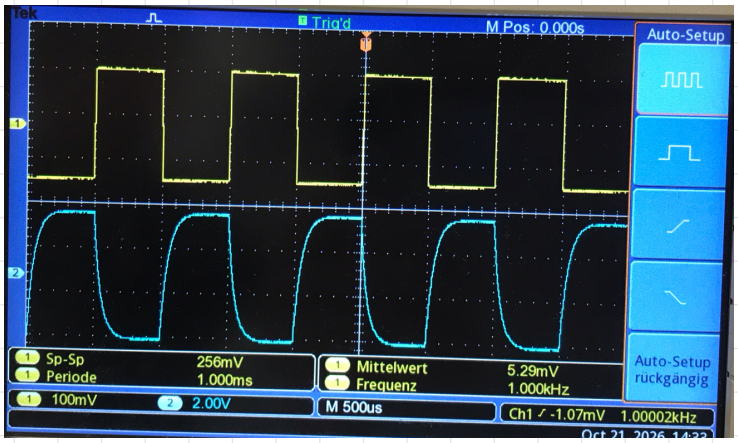

Abb. 3.4

$V = 0{,}3\,V_{SS}$

S1 : 2

S2 : 1

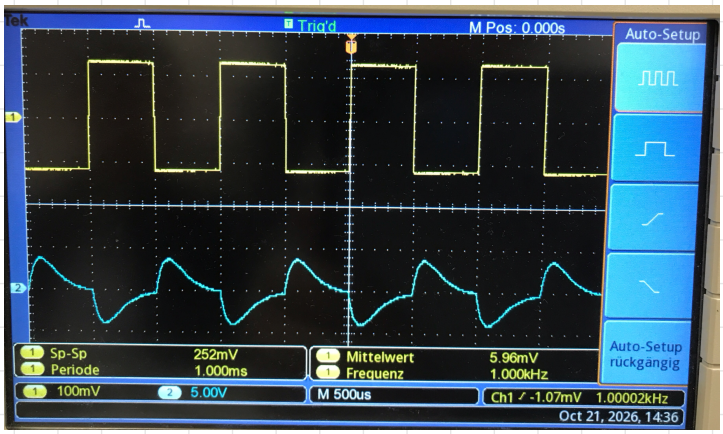

Abb. 3.5

$V = 0{,}3\,V_{SS}$

S1 : 3

S2 : 1

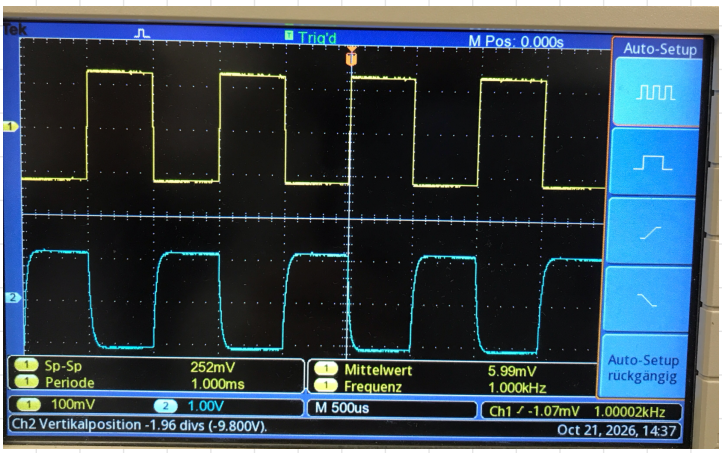

Abb. 3.6

$V = 0{,}3\,V_{SS}$

S1 : 2

S2 : 2

Abb. 3.7

$V = 0{,}3\,V_{SS}$

S1 : 3

S2 : 2

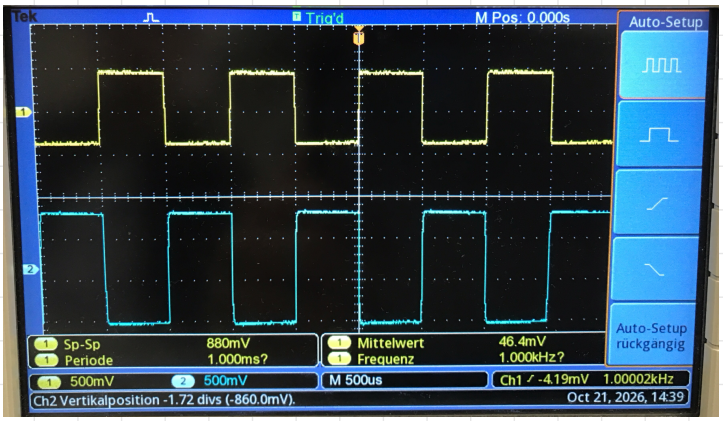

Abb. 3.8

U = 1 Vss

S1 : 2

S2 : 3

Abb. 3.9

U = 1 Vss

S1 : 3

S2 : 3

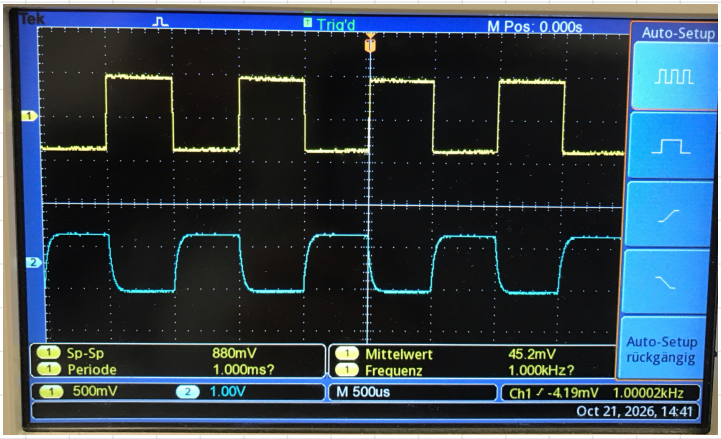

Abb. 3.10

U = 1 Vss

S1 : 2

S2 : 4

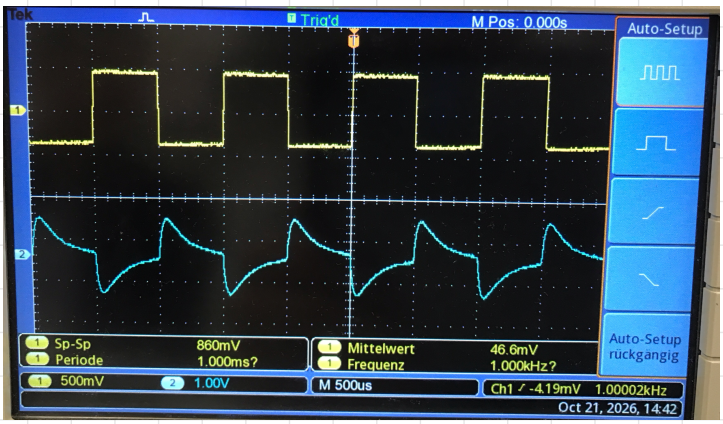

Abb. 3.11

U = 1 Vss

S1 : 3

S2 : 4